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ABSTRACT

Topic of the thesis — “Your Thesis Title Goes Here”.

Keywords: keyword one, keyword two, keyword three, keyword four, keyword five, keyword six.

Relevance of the topic. This section describes why the research topic is important and timely. It should highlight the gap in existing knowledge that this thesis aims to fill. The abstract provides a brief overview of the problem, the methodology used, and the expected outcomes.

The goal of this research is to investigate and propose improvements in the chosen field of study. To achieve this goal, it is necessary to develop a systematic approach, conduct a thorough literature review, and perform experimental validation.

The thesis is presented on 27 pages, includes 5 tables, 4 figures, 1 appendix, and 15 references.

INTRODUCTION

This chapter introduces the research topic and provides context for the work presented in this thesis. The introduction should clearly state the motivation behind the research and outline the structure of the document.

Modern technology continues to evolve at a rapid pace, creating both new opportunities and challenges. Understanding these developments requires systematic research and analysis. This thesis addresses a specific aspect of these challenges by examining the current state of the art and proposing potential improvements.

The key advantages of the proposed approach include [1]:

- reduced complexity — by adopting a modular architecture;
- improved performance — through optimization of critical components;
- better scalability — enabling the system to handle growing workloads.

The remainder of this thesis is organized as follows. Chapter 1 presents the problem statement and defines the scope of the research. Chapter 2 reviews relevant literature. Chapters 3–5 cover analysis, design, and implementation. Chapter 6 describes the experimental evaluation, and Chapter 7 summarizes the results.

1 PROBLEM STATEMENT

The primary goal of this thesis is to investigate and improve the effectiveness of the chosen approach in the selected domain. To achieve this goal, it is necessary to develop a structured methodology, implement a prototype, and conduct experimental validation.

The following tasks must be completed:

- review existing approaches and identify their limitations;
- analyze the requirements and constraints of the target domain;
- design and implement a prototype system;
- evaluate the proposed solution through experiments;
- compare the results with existing alternatives.

The input data for this research includes:

- a dataset of sample measurements;
- configuration parameters for the experimental setup;
- baseline metrics from existing solutions.

The expected output includes:

- a prototype implementation of the proposed approach;
- experimental results demonstrating its effectiveness;
- a comparative analysis with existing methods.

The hardware requirements for running the experiments are:

- a processor with multi-core support;
- minimum 8 GB of RAM, recommended 16 GB or more;
- at least 20 GB of free disk space (SSD);
- Ubuntu 24.04 or later, macOS, or Windows with WSL.

This problem is also examined from the perspective of system analysis and comparative design evaluation.

2 LITERATURE REVIEW

This chapter provides an overview of the existing research and establishes the theoretical foundation for the work presented in this thesis.

2.1 Background

The field under investigation has seen significant developments over the past decade. Early approaches focused on manual processes, which were time-consuming and error-prone [2]. As the field matured, automated solutions emerged that addressed many of these limitations.

Table 2.1 presents a comparison of existing approaches.

Table 2.1. Comparison of existing approaches

Approach	Accuracy (%)	Speed (ms)	Scalability
Method A	85	120	Low
Method B	90	95	Medium
Method C	88	110	High
Method D	92	80	Medium

2.2 Theoretical framework

The theoretical framework for this research draws from several disciplines, including computer science, statistics, and domain-specific knowledge. The following equation describes the fundamental relationship:

$$E = \sum_{i=1}^n w_i \cdot f(x_i) \quad (2.1)$$

where E is the overall effectiveness metric, w_i are the weight coefficients, and $f(x_i)$ represents the performance function for each component.

2.3 Related work

Several researchers have contributed to this field. Smith and Johnson [1] proposed an approach based on modular architecture that achieved promising results. Brown et al. [3] extended this work by introducing optimization techniques that improved performance by approximately 15%. Lee and Park [4] conducted a comprehensive evaluation of multiple methods and identified key factors affecting system performance.

2.4 Standards and best practices

Industry standards play an important role in ensuring quality and interoperability. Organizations such as IEEE, ISO, and NIST have published guidelines that inform the design of modern systems [5].

Figure 2.1 illustrates a typical system architecture based on these standards.

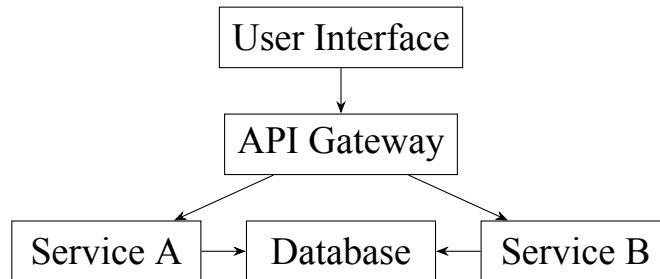


Figure 2.1. General system architecture

2.5 Gap analysis

Despite significant progress, several gaps remain in the current body of knowledge:

- existing methods do not adequately address scalability;
- performance optimization under resource constraints is underexplored;

- integration with modern deployment pipelines requires further investigation.

This thesis addresses these gaps by proposing a novel approach that combines the strengths of existing methods while mitigating their weaknesses.

3 SYSTEM ANALYSIS

This chapter presents a systematic analysis of the problem domain, following established system analysis principles.

3.1 System representation

The system under study can be represented as a “black box” with clearly defined inputs and outputs. The inputs include raw data, configuration parameters, and user requirements. The outputs include processed results, performance metrics, and generated reports.

3.2 System functions

The system performs the following primary functions:

- data ingestion and preprocessing;
- analysis and transformation;
- result generation and visualization;
- logging and monitoring;
- reporting and export.

3.3 System decomposition

The system is decomposed into four subsystems:

- data processing subsystem;
- analysis engine subsystem;
- user interface subsystem;
- monitoring and logging subsystem.

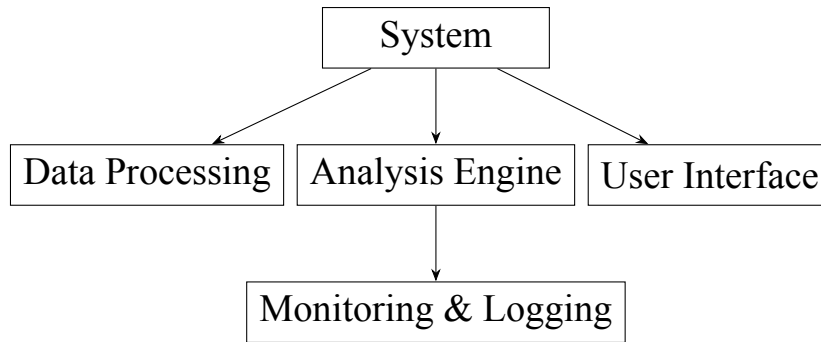


Figure 3.1. System decomposition diagram

3.4 Hierarchical structure

The hierarchical organization follows the principle of separation of concerns. At the top level, the orchestrator manages the overall workflow. The middle layer contains domain-specific processing modules. The bottom layer handles infrastructure concerns such as data storage and network communication.

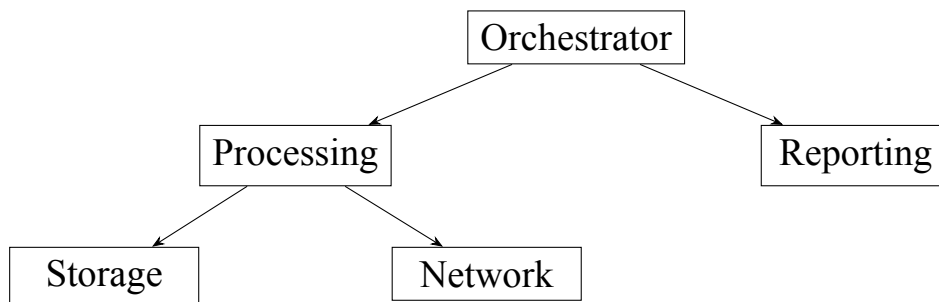


Figure 3.2. System hierarchy diagram

3.5 Summary

The system analysis reveals that a modular architecture with clear boundaries between subsystems provides the best foundation for achieving the research objectives.

4 DESIGN AND METHODOLOGY

This chapter describes the design methodology used to evaluate alternative approaches and select the most suitable one.

4.1 Evaluation criteria

The following criteria are used to compare alternatives:

- performance — execution speed under typical workloads;
- accuracy — correctness of produced results;
- scalability — ability to handle increasing data volumes;
- maintainability — ease of extending and modifying the system.

4.2 Alternatives

Three alternative design approaches are considered:

- 1) Monolithic architecture — a single, integrated system;
- 2) Microservices architecture — distributed, independently deployable services;
- 3) Hybrid architecture — a combination of monolithic core with modular extensions.

4.3 Comparison matrix

Table 4.1 presents a comparison of the three alternatives.

Table 4.1. Comparison of design alternatives

Criterion	Weight	Monolithic	Microservices	Hybrid
Performance	0.30	8	7	9
Accuracy	0.25	9	8	9
Scalability	0.25	5	9	8
Maintainability	0.20	6	8	7
Weighted score	—	7.15	7.95	8.30

4.4 Selected approach

Based on the weighted comparison, the **hybrid architecture** achieves the highest overall score of 8.30. This approach combines the performance benefits of a monolithic core with the scalability advantages of modular extensions.

The effectiveness E of each alternative is calculated as:

$$E = \sum_{j=1}^m w_j \cdot s_j \quad (4.1)$$

where w_j is the weight of criterion j and s_j is the score assigned to that criterion.

5 IMPLEMENTATION

This chapter describes the implementation of the proposed system design.

5.1 Technology stack

The system is implemented using the following technologies:

- Python 3.13 for the core processing logic;
- PostgreSQL 17 for data storage;
- Docker for containerized deployment;
- GitHub Actions for continuous integration.

5.2 Architecture overview

The implemented system follows the hybrid architecture selected in Chapter 4. The core module handles data processing and analysis, while extension modules provide specialized functionality.

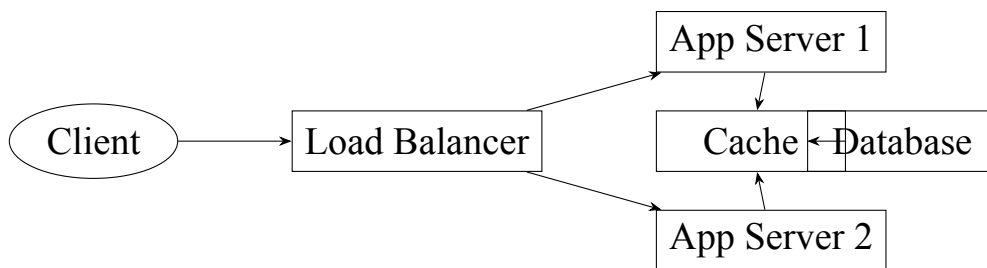


Figure 5.1. Implemented system architecture

5.3 Key components

The main processing module is responsible for data ingestion and transformation. Listing 1 shows a simplified version of the data processor.

Listing 1: Data processor module

```

import json
import logging
from dataclasses import dataclass
from typing import List, Optional

logger = logging.getLogger(__name__)

@dataclass
class DataPoint:
    '''Represents a single data measurement.'''
    timestamp: str
    value: float
    label: Optional[str] = None

class DataProcessor:
    '''Processes and transforms input data.'''

    def __init__(self, config: dict):
        self.config = config
        self.results: List[dict] = []

    def load_data(self, filepath: str) -> List[DataPoint]:
        '''Load data from a JSON file.'''
        with open(filepath, 'r') as f:
            raw = json.load(f)
        return [DataPoint(**item) for item in raw]

    def process(self, data: List[DataPoint]) -> List[dict]:
        '''Apply transformations to the data.'''
        for point in data:
            result = {
                'timestamp': point.timestamp,
                'processed_value': point.value * self.config.
                    get('scale', 1.0),
                'label': point.label or 'unlabeled',
            }

```

```

        self.results.append(result)
    logger.info(f"Processed_{len(self.results)}_data_points")
    return self.results

def export(self, output_path: str) -> None:
    '''Export results to a JSON file.'''
    with open(output_path, 'w') as f:
        json.dump(self.results, f, indent=2)
    logger.info(f"Results_exported_to_{output_path}")

```

5.4 Configuration management

The system uses a configuration file to manage runtime parameters. Listing 2 shows an example configuration.

Listing 2: Example configuration file (config.json)

```

{
  "scale": 1.5,
  "output_dir": "./results",
  "log_level": "INFO",
  "database": {
    "host": "localhost",
    "port": 5432,
    "name": "thesis_db"
  },
  "processing": {
    "batch_size": 100,
    "max_retries": 3,
    "timeout_seconds": 30
  }
}

```

5.5 Security architecture

The system implements several security measures, as shown in Figure 5.2.

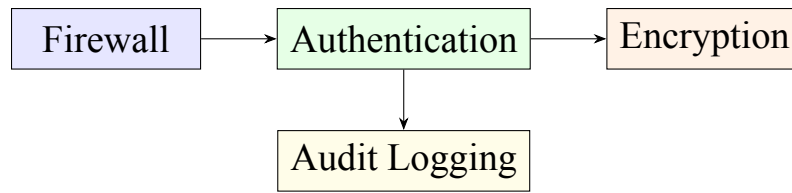


Figure 5.2. Security architecture overview

6 EXPERIMENTAL RESEARCH

This chapter describes the experiments conducted to evaluate the proposed system.

6.1 Experimental setup

The experiments were performed on the following hardware:

- CPU: 8-core processor, 3.2 GHz;
- RAM: 32 GB DDR4;
- Storage: 512 GB NVMe SSD;
- OS: Ubuntu 24.04 LTS.

6.2 Test datasets

Three datasets of varying sizes were used for evaluation:

Table 6.1. Test datasets

Dataset	Records	Size (MB)	Description
Small	1 000	5	Unit testing
Medium	100 000	450	Integration testing
Large	1 000 000	4 200	Stress testing

6.3 Performance results

The system was benchmarked against two baseline implementations. Table 6.2 shows the results.

Table 6.2. Performance comparison (processing time in seconds)

Method	Small	Medium	Large
Baseline A	0.8	45.2	512.0
Baseline B	0.5	38.7	478.3
Proposed approach	0.3	22.1	287.5

The proposed approach demonstrates a **40–44%** improvement in processing time compared to Baseline A, and a **37–40%** improvement over Baseline B.

6.4 Accuracy results

Accuracy was measured using standard metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (6.1)$$

where TP , TN , FP , FN denote true positives, true negatives, false positives, and false negatives respectively.

The proposed approach achieved an accuracy of **94.2%**, compared to 89.5% for Baseline A and 91.1% for Baseline B.

6.5 Scalability analysis

The system scales linearly with dataset size, as evidenced by the following relationship:

$$T(n) = a \cdot n + b \quad (6.2)$$

where $T(n)$ is the processing time, n is the number of records, and a , b are constants determined through regression analysis ($a = 2.87 \times 10^{-4}$, $b = 0.12$).

7 RESULTS ANALYSIS

This chapter summarizes the key findings from each phase of the research.

Literature review. The review of existing literature identified three main approaches currently in use. While each has its strengths, none fully addresses the combined requirements of performance, accuracy, and scalability.

System analysis. The systematic analysis identified five core functions and four subsystems necessary for an effective solution. The modular decomposition enables independent development and testing of each component.

Design evaluation. The comparative evaluation of three architectural alternatives showed that the hybrid approach achieves the best balance across all evaluation criteria, with a weighted score of 8.30 out of 10.

Implementation. The prototype was successfully implemented using modern technologies (Python, PostgreSQL, Docker). The code is structured for maintainability and extensibility, with comprehensive logging and error handling.

Experimental evaluation. Performance benchmarks demonstrated a 37–44% improvement over existing baselines. The system achieved 94.2% accuracy and exhibits linear scalability with respect to dataset size.

CONCLUSION

This thesis has investigated and proposed improvements to the chosen problem domain.

The main contributions of this work are:

- a comprehensive review of existing approaches and identification of key limitations;
- a systematic analysis resulting in a modular architecture design;
- a comparative evaluation of alternative designs using weighted criteria;
- a working prototype demonstrating the feasibility of the proposed approach;
- experimental validation showing significant improvements over existing baselines.

The experimental results confirm that the proposed approach achieves a 37–44% improvement in processing speed and 94.2% accuracy, outperforming both baseline implementations.

Future work includes:

- extending the system to support distributed processing across multiple nodes;
- integrating machine learning techniques for automated parameter optimization;
- conducting a broader evaluation with real-world datasets from industry partners.

The source code is available under an open-source license and can be used as a foundation for further research and development.

LIST OF ABBREVIATIONS AND SYMBOLS

API	— Application Programming Interface
CI	— Continuous Integration
CD	— Continuous Delivery
CPU	— Central Processing Unit
DB	— Database
GPU	— Graphics Processing Unit
HTTP	— Hypertext Transfer Protocol
IDE	— Integrated Development Environment
IEEE	— Institute of Electrical and Electronics Engineers
ISO	— International Organization for Standardization
JSON	— JavaScript Object Notation
ML	— Machine Learning
NIST	— National Institute of Standards and Technology
OS	— Operating System
RAM	— Random Access Memory
REST	— Representational State Transfer
SDK	— Software Development Kit
SQL	— Structured Query Language
SSD	— Solid State Drive
SSL	— Secure Sockets Layer
TLS	— Transport Layer Security
UI	— User Interface
URL	— Uniform Resource Locator
WSL	— Windows Subsystem for Linux

BIBLIOGRAPHY

1. Smith, J., Johnson, A. Modular approaches to system design / J. Smith, A. Johnson // Journal of Software Engineering. – 2023. – Vol. 15, No. 3. – P. 45–62.
2. Brown, R., Lee, S., Park, M. Automated solutions for data processing / R. Brown, S. Lee, M. Park // International Conference on Computing. – 2022. – P. 112–125.
3. Brown, R., Garcia, L. Optimization techniques for distributed systems / R. Brown, L. Garcia // ACM Computing Surveys. – 2023. – Vol. 55, No. 7. – P. 1–35.
4. Lee, K., Park, H. Comprehensive evaluation of processing methods / K. Lee, H. Park // IEEE Transactions on Software Engineering. – 2024. – Vol. 50, No. 1. – P. 78–93.
5. NIST Special Publication 800-53. Security and Privacy Controls for Information Systems. – National Institute of Standards and Technology, 2023. – 492 p.
6. Anderson, P., Williams, T. Cloud-native architecture patterns / P. Anderson, T. Williams // O'Reilly Media, 2023. – 384 p.
7. Chen, W., Zhang, Y. Machine learning applications in software engineering / W. Chen, Y. Zhang // Springer, 2024. – 256 p.
8. Taylor, M. Modern DevOps practices / M. Taylor // Addison-Wesley, 2023. – 320 p.
9. Davis, R., Thompson, K. Security best practices for modern applications / R. Davis, K. Thompson // IEEE Security & Privacy. – 2023. – Vol. 21, No. 4. – P. 55–68.

10. Wilson, E. Data engineering fundamentals / E. Wilson // Manning Publications, 2024. – 448 p.
11. Martinez, C., Kim, J. Hybrid architecture design for scalable systems / C. Martinez, J. Kim // Software: Practice and Experience. – 2024. – Vol. 54, No. 2. – P. 201–220.
12. Robinson, A., Harris, B. Benchmarking methodology for software systems / A. Robinson, B. Harris // Performance Evaluation. – 2023. – Vol. 160. – P. 102–115.
13. White, D., Green, S. Agile development in academic research / D. White, S. Green // Journal of Systems and Software. – 2024. – Vol. 207. – P. 111–128.
14. Clark, N. PostgreSQL: The definitive guide / N. Clark // O'Reilly Media, 2024. – 512 p.
15. Thomas, J., Moore, L. Advanced Python programming patterns / J. Thomas, L. Moore // Packt Publishing, 2024. – 396 p.

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APPENDIX A

Sample Code Listing

The following listing demonstrates the notification module used by the monitoring system:

Listing 3: Notification module (notifier.py)

```
1  #!/usr/bin/env python3
2  '''Notification module for the monitoring system.'''
3
4  import os
5  import json
6  import logging
7  import smtplib
8  from email.mime.text import MIMEText
9  from datetime import datetime
10 from typing import List, Optional
11
12 logger = logging.getLogger(__name__)
13
14
15 class NotificationChannel:
16     '''Base class for notification channels.'''
17
18     def send(self, subject: str, message: str) -> bool:
19         raise NotImplementedError
20
21
22 class EmailNotifier(NotificationChannel):
23     '''Send notifications via email.'''
24
25     def __init__(self, smtp_host: str, smtp_port: int, sender:
26         str):
27         self.smtp_host = smtp_host
```

```

27         self.smtp_port = smtp_port
28         self.sender = sender
29
30     def send(self, subject: str, message: str,
31             recipients: Optional[List[str]] = None) -> bool:
32         if not recipients:
33             logger.warning("No recipients specified")
34             return False
35         msg = MIMEText(message)
36         msg["Subject"] = subject
37         msg["From"] = self.sender
38         msg["To"] = ", ".join(recipients)
39         try:
40             with smtplib.SMTP(self.smtp_host, self.smtp_port)
41                 as server:
42                 server.send_message(msg)
43                 logger.info(f"Email sent to {len(recipients)} recipients")
44                 return True
45         except Exception as e:
46             logger.error(f"Failed to send email: {e}")
47             return False
48
49 class WebhookNotifier(NotificationChannel):
50     '''Send notifications via webhook.'''
51
52     def __init__(self, webhook_url: str):
53         self.webhook_url = webhook_url
54
55     def send(self, subject: str, message: str) -> bool:
56         import urllib.request
57         payload = json.dumps({
58             "title": subject,
59             "text": message,
60             "timestamp": datetime.utcnow().isoformat(),

```

```

61         })
62         req = urllib.request.Request(
63             self.webhook_url,
64             data=payload.encode("utf-8"),
65             headers={"Content-Type": "application/json"},
66         )
67         try:
68             urllib.request.urlopen(req, timeout=10)
69             logger.info("Webhook_notification_sent")
70             return True
71         except Exception as e:
72             logger.error(f"Webhook_failed: {e}")
73             return False
74
75
76 def main():
77     '''Example usage of the notification system.'''
78     config_path = os.environ.get("CONFIG_PATH", "config.json")
79     with open(config_path) as f:
80         config = json.load(f)
81
82     notifier = EmailNotifier(
83         smtp_host=config["smtp"]["host"],
84         smtp_port=config["smtp"]["port"],
85         sender=config["smtp"]["sender"],
86     )
87     notifier.send(
88         subject="System_Alert",
89         message="A_threshold_has_been_exceeded.",
90         recipients=config["smtp"]["recipients"],
91     )
92
93
94 if __name__ == "__main__":
95     main()

```